

FILTER DESIGN TEST CASE

Lattice Filter - n79 (4.4-5.0 GHz 5G NR) for IoT Narrowband Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Author:	Dr. Raj Patel
Reviewer:	Dr. Anya Krishnamurthy
Design Center:	Andover HQ Lab
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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Lattice Filter filter targeting n79 (4.4-5.0 GHz 5G NR) for IoT Narrowband Module. The filter is designed on PZT on Si substrate with a target center frequency of 5068.3 MHz and bandwidth of 152.2 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the IoT Narrowband Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Lattice Filter
- Frequency Band: n79 (4.4-5.0 GHz 5G NR)
- Application: IoT Narrowband Module
- Substrate: PZT on Si
- Package: SiP (System in Package)
- Design Tool: Ansys HFSS 2024R1
- Design Center: Andover HQ Lab

This specification applies to all variants of the Lattice Filter design targeting n79 (4.4-5.0 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5068.3 MHz
Bandwidth (3 dB):	152.2 MHz
Insertion Loss Target:	≤ 1.3 dB
Return Loss Target:	≥ 20.0 dB
Out-of-Band Rejection:	≥ 39.0 dB
Isolation (Duplexer):	≥ 40.0 dB
Q Factor:	1960
Temperature Coefficient:	-17.0 ppm/°C
Die Size:	2.3 x 0.7 mm
Wafer Size:	6-inch
Number of Resonators:	4
Electrode Material:	Mo

Passivation: SiO2

4. TEST PROCEDURE

1. Open Ansys HFSS 2024R1 and load the Lattice Filter template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 5068.3 MHz, BW 152.2 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, PZT on Si).
10. Perform on-wafer probing using Keysight N5227B PNA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 15205 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.3 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 228 MHz offset >= 39.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 10.8 MHz
- Power handling shall be >= +31 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2024-03-26
Devices Tested: 90
Insertion Loss (meas): 1.02 dB
Return Loss (meas): 22.2 dB
Rejection (meas): 41.8 dB
Isolation (meas): 44.7 dB
Q Factor (meas): 1960
Group Delay Variation: 3.6 ns
Power Handling: +30 dBm
Die Yield: 86.5%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Lattice Filter design demonstrated excellent performance for n79 (4.4-5.0 GHz 5G NR).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Group delay flatness improved compared to previous revision.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

Author: Dr. Raj Patel Date: 2024-03-26
Reviewer: Dr. Anya Krishnamurthy Date: 2024-03-27
Approver: Dr. Yuko Ishida Date: 2024-04-04

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